

Facilitator answer key — Holocron capstone worked solution.

Facilitator-only — lives in the facilitator folder of every repo; do not hand to participants. Part of the internally-consistent set in this folder (see `README.md`).

AI Spec — Holocron (release-1 slice)

Sibling to: PRD-light, TCD-light, TMD-light, SEP-light, DP-light |

Status: Reviewed **AI Provenance:** AI-assisted draft; every section validated against a source artifact — see §8.

The integrated, engineering-ready spec — deliberately *shorter* than the sum of PRD + TCD + TMD, with citations back to the deeper artifacts.

1. Headline

Holocron gives content owners across 150+ countries one governed place to create, review, and publish UI strings — so 20–30+ apps fetch localized text at render ($\leq 200\text{ms}$ p95) without hardcoding or a redeploy.

2. Engineering-ready summary

- **Problem + users (PRD §1–3).** UI text is hardcoded inside each app, so any change needs an engineering change + a redeploy across 20–30+ apps. Holocron centralizes strings in a governed store. Users: *content owners* (create/edit/publish `en-US` + translation variants), *reviewers* (request/complete review), *consuming apps* (read at render). Scale: ~500k strings, 30–40 locales, `en-US` source, Internal data.
- **Architecture (TCD §1).** A modular service on AKS with two paths over one PostgreSQL system of record: a low-traffic authoring path (management app 99.5%) and a read-optimized delivery path fronted by **Redis + API Management/App Gateway** (99.9%, p95 $\leq 200\text{ms}$). Publish + audit-write stay in one transaction.
- **Data + delivery (TMD §1, §3).** Immutable by design: namespaced **SourceString** key, immutable **versions** for source + translations, a **PublicationSelection** (approved | previous | `en-US` fallback) per publish, an immutable **AuditEntry** per mutating op. Apps read `GET .../namespaces/{ns}?locale=...` (Entra token) and get published strings with an explicit `fallback` flag.
- **How to build it.** Authoring CRUD + review against Postgres with an append-only audit stream (Event Hubs → Blob); delivery as a separate horizontally-scaled module reading through Redis (read-through + backfill,

degrades to direct Postgres). Publish invalidates the cache — ≤ 60 s normal, synchronous < 5 s for critical/legal. Identity, RBAC, and rate limiting are enforced at the APIM edge and re-checked server-side.

3. Data + API contract

Entities (TMD §1), versions append-only: Product · SourceString (immutable namespaced key `product.namespace.key`, `criticality`) · SourceStringVersion · TranslationVariant · TranslationVersion · ReviewRequest · PublicationSelection · AuditEntry · User (Entra ID; roles Authenticated Employee / Content Owner / Reviewer / Review Admin / Holocron Admin — PRD §7).

Delivery — the read contract:

```
GET /v1/products/{productId}/namespaces/{namespace}?
  locale=<BCP-47> [Bearer Entra token]
200 → [ {key,value,locale,version,fallback}, ... ]
401 no/invalid token · 403 wrong product scope · 404
unknown ns · 429 over 50 req/s (burst 100)
```

`fallback:true` = no published translation for the locale; `en-US` served instead.

Authoring — RBAC-gated:

```
POST .../namespaces/{ns}/strings          create
string (+ v1) [Content Owner]
POST .../strings/{key}/versions           edit = new
version [Content Owner]
POST .../strings/{key}/review-requests    request
review [Content Owner]
PATCH .../review-requests/{id} {complete} complete
review [Reviewer]
POST .../strings/{key}/publish            publish (+
AuditEntry) [Content Owner / Holocron Admin]
normal: publish without review allowed, as an
audited override.
critical/legal: completed compliance review
REQUIRED (else 409) + synchronous <5s invalidation.
```

4. Sequence + failure handling

```
HAPPY: publish → Postgres write version + AuditEntry
(one tx) → Redis invalidate → Event Hubs (async)
consumer GET → Entra OK at APIM → Redis HIT →
200 [fallback:false] p95  $\leq 200$ ms; visible  $\leq 60$ s
SAD: missing translation → serve en-US,
fallback:true (never 404 a live key)
cache miss → read-through Postgres +
backfill; Redis down → direct Postgres reads
401/403/404; 429 over 50 req/s → consumer
holds last good cache
WEIRD: Event Hubs down → buffer + retry; the
Postgres publish still commits (audit never blocks
the write)
```

5. Constraints

Constraint	Target
Delivery latency	p95 ≤ 200ms / p99 ≤ 500ms
Delivery / management availability	99.9% / 99.5%
Publish → delivery propagation	≤ 60s normal · <5s critical/legal
Rate limit	50 req/s, burst 100

Security (TCD §3–4, PRD §7): Entra ID token at the APIM edge + server-side; RBAC per operation (product-scoped, least-privilege delivery identity); immutable AuditEntry per op → Event Hubs → Blob; secrets in Key Vault; auth fails closed.

Compliance: the audit trail is the control for the optional-review default on normal strings; critical/legal add an enforced pre-publish review + <5s invalidation (SEP §3). **Scale:** delivery pods scale on read QPS and survive a management outage via Redis + a Postgres read replica.

6. Decisions

Made: (1) **Redis read-cache**, ≤60s staleness for normal strings; critical/legal get synchronous <5s (SEP §3). (2) **Reviews optional + audited override** for normal strings; critical/legal require enforced pre-publish review (SEP §3). (3) **Immutable namespaced keys + versions** (rename = create-new + deprecate). (4) **Missing translation** → en-US **fallback** with an explicit indicator (PRD §5 / CS7 AC), so a live key never 404s. **Not decided (open):** RPO/RTO for Postgres + audit archive (→ Platform/SRE); alias governance (out of slice); export/import format (open).

7. Stakeholders + sign-off (SEP §1 + TCD §6)

Stakeholder	Interest	Status
Anna Woods — Product	Scope slice + R1 value	Approved
Nick Grant — Engineering	AKS, delivery SLOs, Redis/Postgres split	Approved
Enterprise Infrastructure	Entra ID + AKS + secrets	Approved (pending Key Vault rotation runbook)
Legal / Compliance	Audit + review gates	Approved — critical/legal flag enforces pre-publish review + <5s (SEP §3)
Architecture	Immutable namespaced key schema	Approved

8. Provenance log

AI-drafted; human judgment shaped every section. **A spec that ships without this section ships fiction.**

#	Section	Validation	Status
1	Headline	Scale + p95 vs PRD §1–3 / TCD §3; roles vs PRD §7	✓
2	Eng summary	Each claim traced to PRD §1–5, TCD §1, TMD §1/ §3; IN/OUT vs PRD §4	✓
3	Data + API	Entities + immutability vs TMD §1; endpoint/codes/ <code>fallback</code> vs TMD §3; roles match PRD §7	✓
4	Sequence	Vs TCD §2 failure handling + TMD §4; SLO figures match TCD §3/§4	✓
5	Constraints	Every SLO copied (not paraphrased) from TCD §3/§4; security vs STRIDE + PRD §7	✓
6	Decisions	3 map to TCD §5 trade-offs; en-US fallback from PRD §5 / CS7 AC; open items confirmed not-decided	✓
7	Sign-off	Names from SEP §1; Legal resolved via the flag (SEP §3); Key Vault condition on Enterprise Infrastructure	✓
8	This log	Instructor confirms every row cites a real source before this spec is "Reviewed"	✓